

Introduction to the *Principia*

Isaac Newton (1642–1727) is one of the most influential scientists and mathematicians of all time, and a key figure in the scientific revolution. He is considered one of the two or three best theoretical physicists and among the very best experimental physicists and mathematicians in history. His celebrated book *Mathematical Principles of Natural Philosophy* (we will use its abbreviated Latin name *Principia*), first published in 1687, is the most important scientific book ever written, and his discoveries are still the basis for much scientific and mathematical research.

A generation before Newton, Galileo invented the experimental part of the scientific method when he devised experiments to answer specific questions about the nature of the universe. He set the stage for Newton by declaring the universe is a “book” that is “written in the language of mathematics” and using math to describe the results of his experiments. Newton invented the theoretical part of the scientific method when he devised his laws of motion and gravity, and used them to predict the answers to specific questions about the nature of the physical universe.

The *Principia* is Newton’s masterful rendition of those questions and the methods he used to answer them. His answers are wonderful and often dazzling, but when viewed in the historical and scientific context of the late 17th century, his *questions* are truly extraordinary.

Newton’s geometric methods and techniques, as well as the geometry and vocabulary that passed for common knowledge in the late 17th century, are now arcane and all but inaccessible to a modern reader. The contents of the *Principia* are not. Our goal is to present all the wonderful physics in the *Principia* – those extraordinary questions and answers – in a manner that a modern reader can recognize and understand using 21st century physics and mathematics.

After a brief review of the events in Newton’s life that led to the *Principia* and a short summary of its logical structure, this INTRODUCTION will cover two important topics. First we summarize the physics and math you the modern reader need to work your way through our rendition of Newton’s work. Second we present a summary of the state of physics in the mid-17th century when Newton began his research on mechanics and gravity.

Then we will be ready to start our way through the *Principia*!

A Brief History* of the *Principia*

The publication of the first edition of the *Principia* in 1687 was the culmination of over 20 years of intermittent work on mechanics and gravitation that Newton started while he was still an undergraduate. Newton enrolled at Trinity College of Cambridge University in 1661 when he was 18, and by his third year he had abandoned the stagnant classical curriculum and started his own original research program in mathematics and physics. He received his Bachelor of Arts degree in August 1665, just before the plague outbreak forced the university to close.

Newton retreated to his family home at Woolsthorpe and he did not return to Cambridge permanently until April 1667. This is Newton's so-called "miracle year" when he continued his work on mathematics, mechanics, and optics. It is also when the famous "falling apple" incident allegedly happened. [1]

By 1669 Newton had developed three approximations that he later used to great effect doing his orbital calculations in the *Principia*. In October 1669, less than five years after receiving his B.A. degree, Newton became Lucasian Professor of Mathematics at Cambridge University.

By the early 1670s, his research centered on optics, alchemy, and theology, and in 1672 he submitted material on optics to the Royal Society that they subsequently published in their *Philosophical Transactions*. Robert Hooke and Christiaan Huygens objected to several parts of Newton's optical theory, and the subsequent exchanges strongly affected the thin-skinned Newton, who completely withdrew from public discourse on scientific topics until after the *Principia* was published.

Newton was concentrating on chemistry and alchemy when Hooke sent him one of the most consequential letters in the history of physics. The subsequent correspondence stimulated Newton to prove an elliptical planetary orbit results from an inverse-square attractive central force. Between November 1679 and January 1680, Hooke and Newton exchanged 6 letters (4 from Hooke, 2 from Newton). [10]

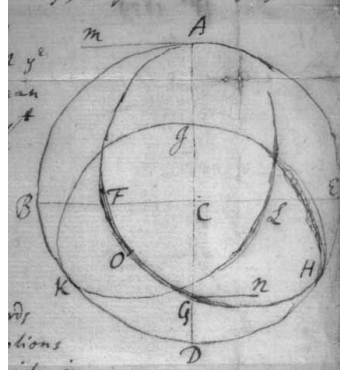
If we look on the publication of the *Principia* as the birth of physics, the seminal moment may be Hooke's first letter (dated November 24, 1679) when he asked Newton to give an opinion on Hooke's idea to resolve the celestial motion of planets into two components: inertial, straight-line motion in the tangential direction and an attractive accelerated motion towards the central body.

Newton claimed not to have heard of Hooke's idea and admitted his research interests had moved away from mechanics. But Hooke's letter had temporarily rekindled Newton's interest, and in his second letter to Hooke (dated December 13, 1679), Newton included a sketch of an orbit through a non-resisting medium for a constant central force.

* For a better-written, not-so-brief history, see Chapter 1 of Cohen's Guide. [4]

Newton's orbit starts at a radial maximum at point A , moves counterclockwise about the center at point C , and first crosses the horizontal axis at point F . It reaches its first radial minimum at point O , the next maximum is at point H , and a crude measurement shows the apsidal angle $\angle OCH$ is approximately 108° .

Point	Description
A	start of orbit
C	center of orbit
F	first x -axis crossing
O	minimum distance
G	first y -axis crossing
H	maximum distance
$\angle OCH$	apsidal angle



Newton's sketch of an orbit for a constant central force. [17]

The orbit Newton drew closely resembles a periodic orbit with an apsidal angle of $\Phi_A = \frac{2}{3}\pi = 120^\circ$ but no such period-2 orbits exist for a constant central force. Periodic orbits for the central force $f(r) = \beta r^n$ are found in families of orbits with the same apsidal angle that start at particular bifurcating exponents. No family of period-2 orbits contains the constant force ($n = 0$). [18]

The family with the smallest periodicity that includes the constant force is a family of period-4 orbits. There is a period-4 constant-force orbit with $\Phi_A = \frac{4}{7}\pi \approx 102.9^\circ$ and angular momentum $\ell = 0.565 \ell_{\text{cir}}$ that is very similar to Newton's orbit. There is a period-2 orbit for $n = -0.80585$ and $\ell = 0.629 \ell_{\text{cir}}$ with $\Phi_A = 120^\circ$ that is also very similar to Newton's orbit.

In the last letter of their exchange (dated January 17, 1680), Hooke implored Newton to address the problem of finding the orbit under an inverse-square attractive force. [19] Newton did not continue working on orbital mechanics after the correspondence with Hooke ended, except when the comet of late-1680/early-1681 interested him enough to attempt to calculate its trajectory from observations he collected. Between 1680 and 1684, Newton worked mostly on alchemy and theology, but Hooke had planted a seed that needed one more event to germinate fully.

That now-famous event was Edmond Halley's visit to Newton in August 1684. In January 1684, Halley, Hooke and Christopher Wren engaged in a coffee-shop discussion about the connection between elliptical orbits and the inverse-square force. Halley admitted he could not prove it and Hooke claimed he had a proof that he wanted to keep private. A skeptical Wren offered a valuable book as a prize to the first person to prove within two months that the inverse-square force law leads to elliptical orbits.

Six months later, Halley was traveling on family business and decided to visit Newton, who by then was well-known to the Royal Society. During this visit, Halley asked Newton a question about inverse-square orbits: What orbit results for an attractive inverse-square force? When Newton defended his answer “it is an ellipse” by saying “I have calculated it” a surprised Halley asked to see the calculation. Newton claimed he was unable to find it but promised to redo the calculation and send it to Halley.

Maybe his comment to Halley meant he had already done the calculation Hooke suggested for the inverse-square force, maybe he had already done it but wanted to check it, or maybe it simply reflected his confidence that he could do it. The physically impossible orbit he drew in the letter to Hooke is not evidence to the contrary. He made a mistake drawing the orbit but his calculation (which probably used his curvature method) was accurate. [20]

Not one to repeat a mistake, Newton returns to constant-force orbits with a correct, more rigorous analysis in Example 1 of PROPOSITION 45 in BOOK I of the *Principia* (see page 159 in SECTION 9).

There is some question about Halley’s question: It is not clear if Halley asked Newton the orbit-to-force question (what gravitational force produces an off-center elliptical orbit) or the force-to-orbit question (what orbit results for an inverse-square gravitational force). It does not really matter: Newton answered both questions.

Newton kept his promise and his answer arrived at the Royal Society in November 1684, just a few months after Halley’s visit, in the form of a 9-page monograph called *De motu corporum in gyrum* (*On the motion of bodies in an orbit*). This short document contains eleven propositions where Newton introduced the term “centripetal” force, presented a theoretical derivation of what we now call Kepler’s Laws, discussed uniform circular motion, and solved both the orbit-to-force and force-to-orbit problems for elliptical orbits and the inverse-square force. Newton ultimately included all of these propositions in the *Principia*.

This incredible document was the core that Newton developed and expanded into the 500-page *Principia*. Newton temporarily abandoned his research on alchemy and theology, and attacked the problem of orbits and gravity with all his intellectual might. After 18 months of intense research and writing by Newton, and Halley’s intellectual, emotional, and financial support, the first edition of the *Principia* was published on July 5, 1687. It made the world aware that Newton was its foremost physicist and mathematician. Newton was 44 years old.

Newton was a genius of the highest order and deserves most of the glowing accolades we bestow on him. But several myths persist about Newton’s life and work that distort and often diminish his legacy.

The Miracle Year

Myth: Newton's accomplishments came in a flash of insight as fully-formed theories that were complete when he returned to Cambridge.

It took Newton two decades to develop the full theory of Universal Gravitation he published in the *Principia* in 1687. In a 1669 manuscript where he derived mv^2/r (more than a decade after Huygens), Newton was still thinking of circular motion in terms of a centrifugal tendency away from the center and not a centripetal force towards the center. He makes no mention of an attractive gravitational force to explain the planetary orbits.

The historical record shows that Newton wrote most of his mathematical papers on calculus while he was at Cambridge. He made his first major discovery – the generalized binomial theorem – in 1665 before he left Cambridge. He was extremely productive during his time at Woolsthorpe and he made significant progress in three separate fields (calculus, gravity, and optics) but nothing unique happened there during the so-called miracle year. While he was there, Newton simply followed the research program that he started as an undergraduate in 1664 and continued when he returned to Cambridge in 1667. [21]

The Falling Apple

Myth: The flash of insight was the result of Newton being bonked on the head by a falling apple in the summer of 1666.

Newton himself may be the original source of this myth. [1, page 154]

In April 1726, some 60 years after the alleged event, Newton shared the apple anecdote with William Stukeley, who included it in his 1752 biography of Newton.

“After dinner, the weather being warm, we went into the garden and drank tea under the shade of some apple trees. He told me, he was just in the same situation, as when formerly, the notion of gravitation came into his mind, occasion'd by the fall of an apple, as he sat in a contemplative mood.”

Newton probably embellished this story to support his argument against Hooke's accusation of plagiarism over the inverse-square law of gravity.

The apple tree in front of Newton's house is still there and it is easy to believe Newton saw an apple fall from it. Whether the anecdote was the fanciful imaginings of an old man with an agenda or an accurate recollection of an inspirational event, the legend of the falling apple is now firmly ensconced in our culture as the second great “eureka” moment in the history of science. The story mistakenly compresses decades of work into a single inspirational moment. Perhaps we are responsible for the myth's popularity and persistence, for it offers us the delusion that we too can experience genius if only the right apple hits us at the right time. If only...

Calculus and the Principia

Myth: Newton invented calculus in 1666, used it to do the calculations in the *Principia*, and then translated his results into the geometrical method he published.

There is no evidence to support this frequently repeated myth. [22, 23]

To bolster his claim in the conflict with Leibniz over the priority of inventing the calculus, Newton himself often claimed he originally proved many propositions in the *Principia* analytically using calculus. Much of the mathematics Newton used in the *Principia* involved a geometric synthesis of limits, and he did occasionally use his “method of fluxions” and binomial expansions. Not a single document has ever been found that shows Newton doing calculations in private using methods that differ from the version he published in the *Principia*. [4, see §5.8]

The Structure of the *Principia*

The third and final edition of the *Principia*, published in 1726 just a year before Newton’s death, contains four Prefaces, a section of DEFINITIONS, Newton’s three LAWS OF MOTION, three “BOOKS” that make up the main body of work, and a concluding GENERAL SCHOLIUM.

Author’s Preface to the Reader

In his Author’s Preface to the 1st edition, Newton announces the main topic of the *Principia* is “the science, expressed in exact propositions and demonstrations, of the motions that result from any forces whatsoever and of the forces that are required for any motions whatsoever”. For Newton, the “basic problem” that physical science addresses is “to discover the forces of nature” by observing motion and then to predict what other motions arise from those forces.

Newton explicitly states his main goal for the *Principia*, and in the process defines the fundamental problem of celestial mechanics: First solve the **orbit-to-force problem** – find the central force that produces the observed orbits – and then once the force is known, solve the **force-to-orbit problem** and predict what other motions are possible for that force. We can think of the *Principia* as a study of various forces and the motions they produce. Newton called this “rational mechanics” in his Preface to the 1st edition, and later his adversary Gottfried Leibniz dubbed it “dynamics” (much to Newton’s chagrin). [4]

Newton added a short Author’s Preface to each of the 2nd and 3rd editions where he pointed out the changes in the newer editions. In the 2nd edition, his editor Roger Cotes added a long Editor’s Preface where he strongly defended Newton’s theory of universal gravitation and argued against the mechanical vortex theory of René Descartes.

Definitions

Newton begins the *Principia* with eight definitions and a scholium about the absolute nature of space and time. (A **scholium** is a note added to a text as an explanation, criticism, or commentary.) He wants to avoid any confusion his new technical definitions of generally common words might cause. The modern names for the eight terms he defines are Mass, Momentum, Inertia, Applied Force, Central Force, Force Constant, Acceleration, and Net Force.

Newton's definition of what we call a central force includes his first published use of the term "**centripetal**" force he had recently invented. The word **centripetal** means "moving or tending to move toward a center" and it comes from the words for "center" and "seeking" in Latin. A modern reader often associates circular orbits with the word "centripetal" but Newton used it more generally to mean *any attractive central force* and that is the phrase we will use.

In the scholium, Newton explains his views on position and motion, time and space. He makes a distinction between the values we measure relative to real physical objects and the "absolute, true, and mathematical" nature of time and space. Newton argues that the distinction between their true nature and their relative measured values is necessary and justified.

Laws of Motion

Newton's next topic is his three LAWS OF MOTION, which he also calls axioms. An **axiom** is a statement which is accepted without proof as self-evidently true. In this section, Newton states his laws of motion, adds six corollaries, and a long scholium where he presents the results of his experiments with colliding pendulums. These laws describe the relationship between any object, the forces acting upon it, and the resulting changes in motion. They are the foundation upon which classical mechanics is built.

The modern versions of these laws are familiar to everyone who studies physics.

- **Newton's 1st Law:** The Inertia Law $\vec{F} = 0 \iff \Delta\vec{v} = 0$
- **Newton's 2nd Law:** The Acceleration Law $\vec{F} = m\vec{a} = \frac{d\vec{p}}{dt}$
- **Newton's 3rd Law:** The Reaction Law $\vec{F}_{12} = -\vec{F}_{21}$

Newton's statement of the 2nd Law in the *Principia* is not the modern version we use and teach today. He was thinking about both impulses and continuous forces when he said the change in motion is proportional to the applied force with no mention of rate.

Three BOOKS

Newton divides the rest of the *Principia* into three parts that he calls BOOKS. In BOOK I (which he calls “THE MOTION OF BODIES”) Newton develops a mathematical theory of motion under the influence of various attractive central forces. Newton gives BOOK II the same name, but “MOTION IN RESISTING MEDIA” is more descriptive. In BOOK II he studies motion affected by resisting forces (such as air or water resistance) in an attempt to understand his own experiments on free fall and pendulums, and to refute the Cartesian idea that the planets are carried around their orbits by fluid vortices.

In BOOK III Newton shows us “THE SYSTEM OF THE WORLD” where he applies what he has learned in the first two books to deduce the gravitational force that all objects exert on each other. He then uses that force to calculate the motions of the planets, the Moon, the oceans, and comets.

Newton divides the first two books into sections that contain **propositions**, **corollaries**, **lemmas**, and **scholiums** on a particular topic. Each proposition is either a theorem to be proved or a problem to be solved. A **corollary** is a statement that follows with little or no proof from the previously proven proposition. In the *Principia*, a **lemma** is usually a mathematical proposition that Newton needs to prove a theorem or solve a problem. As we saw earlier, a **scholium** is an explanatory note added to a text as a commentary.

BOOK I has fourteen sections that contain 98 propositions, 29 lemmas, and several scholiums. BOOK II has nine sections containing 53 propositions, 7 lemmas, 17 short scholiums, and a long GENERAL SCHOLIUM where Newton reports on the results of seven pendulum experiments where he measured the arc lost in an attempt to determine the speed dependence of the resisting force.

Newton organizes THE SYSTEM OF THE WORLD of BOOK III differently. It opens with two short sections: The first contains four RULES FOR THE STUDY OF SCIENCE, and the second has six observed PHENOMENA that Newton will use to derive Universal Gravitation. The third section contains 42 propositions, 11 lemmas, 6 short scholiums, and 2 hypotheses; the final section is the concluding GENERAL SCHOLIUM where Newton makes his famous “*hypotheses non fingo*” statement.

Newton does not separate the propositions in BOOK III but we categorize them into six topics: PROPOSITIONS 1-9 present his derivation of the Law of Universal Gravitation. PROPOSITIONS 10-20 explore the implications and consequences of this law for orbiting and rotating objects. PROPOSITIONS 21-35 explain the various motions of our Moon. PROPOSITIONS 36-38 explore the forces on Earth’s oceans and the shape of the Moon. PROPOSITION 39 is Newton’s explanation of the precession of the equinoxes, and PROPOSITIONS 40-42 are his solution for determining comet trajectories from Earth-bound observations.

Modern Preliminaries: What We Need to Know

Our exploration of the physics in the *Principia* will be in the modern language of calculus and vectors as understood by physicists and mathematicians in the 21st century. The standard curriculum we teach current physics students covers almost all the physics and math we need to understand the *Principia*, and much of it is at the undergraduate level. These modern preliminaries are the minimum we need to know to understand the *Principia*.

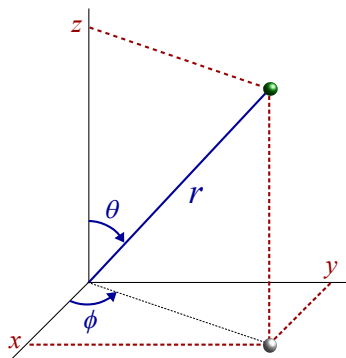
We start with a summary of the **notation** we will use, a brief review of **vectors**, and the expressions for **velocity** and **acceleration** in various coordinate systems. We will use **Newton's 2nd Law** extensively, so we review methods for solving it in its modern form. We finish with two other important topics — **central forces** and **energy conservation** — we will use frequently.

Mathematical Notation

We use standard physics notation for most of our expressions and definitions. In some cases however, it seems appropriate to use the notation favored by astronomers even though it may seem strange to physicists. For example, we follow the standard astronomical use of ω to denote the periapsis angle between an orbit's periapsis and its ascending node. (The **periapsis** is the point on an orbit where the orbiting object is nearest to the central object it orbits.)

The dot notation $\dot{r}$ indicates a time derivative and double-dot $\ddot{r}$ means the second time derivative. The prime notation usually indicates a spatial derivative (although we will occasionally use r' to denote a final radial distance) and double-prime r'' means the second spatial derivative. Small differences δr are not infinitesimal, and infinitesimal differences dr are used “in the limit” in the usual calculus sense.

Spherical Coordinates	
$x = r \sin \theta \cos \phi$	$r = \sqrt{x^2 + y^2 + z^2}$
$y = r \sin \theta \sin \phi$	$\phi = \arctan y/x$
$z = r \cos \theta$	$\theta = \arccos z/r$
Coordinate Transformations	



We use the physics convention for three-dimensional spherical coordinates where we measure the polar angle θ from the $+z$ axis and we measure the azimuthal angle ϕ from the $+x$ axis in the x - y plane (where $\theta = 90^\circ$ and $z = 0$). The polar angle θ ranges from 0 to π (180°) and the azimuthal angle ϕ ranges from 0 to 2π (360°).

Vectors

Mathematics is a collection of tools that physicists use to construct theories that describe, explain, and predict facts about the physical universe. Vectors are particularly useful and powerful tools. A **vector** is a quantity that has both a magnitude and a direction

In Cartesian coordinates, we describe a two-dimensional vector with its x and y components:

$$\vec{F} = F_x \hat{x} + F_y \hat{y} \quad (\text{V-1})$$

The unit vectors $\hat{x}$ and $\hat{y}$ are constant and they point along the x and y axes respectively. Like all unit vectors, they are dimensionless with unit magnitude.

We can use the components to calculate any vector's magnitude and direction:

$$F = \sqrt{F_x^2 + F_y^2} \quad (\text{V-2a})$$

$$\theta_F = \tan^{-1} \frac{F_y}{F_x} \quad (\text{V-2b})$$

In polar coordinates, we describe a vector with its radial component F_r and its azimuthal component F_ϕ :

$$\vec{F} \equiv F_r \hat{r} + F_\phi \hat{\phi} \quad (\text{V-3})$$

The unit vectors $\hat{r}$ and $\hat{\phi}$ point in the direction of increasing r and ϕ respectively, and they are mutually perpendicular. The radial unit vector $\hat{r}$ points away from the origin so $-\hat{r}$ points toward the origin. The azimuthal unit vector $\hat{\phi}$ points counterclockwise around the origin, and in general $\hat{\phi}$ does *not* point in the direction tangential to the trajectory.

Unlike their Cartesian counterparts, the polar unit vectors are not constant and their directions depend on their location:

$$\hat{r} = +\hat{x} \cos \phi + \hat{y} \sin \phi \quad (\text{V-4a})$$

$$\hat{\phi} = -\hat{x} \sin \phi + \hat{y} \cos \phi \quad (\text{V-4b})$$

A change in the distance r from the origin does not change the direction of the polar unit vectors but a change in the angle ϕ does.

Velocity and Acceleration in Cartesian coordinates

The position vector $\vec{r} = x\hat{x} + y\hat{y}$ points from the origin to the object's location, and its components can be functions of time. The velocity is the rate of change of position:

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt} \hat{x} + \frac{dy}{dt} \hat{y} \quad (\text{A-1})$$

The x -component of the velocity is $v_x = \dot{x}$ and the y -component is $v_y = \dot{y}$. The speed $v = \sqrt{\dot{x}^2 + \dot{y}^2}$ is the magnitude of the velocity vector.

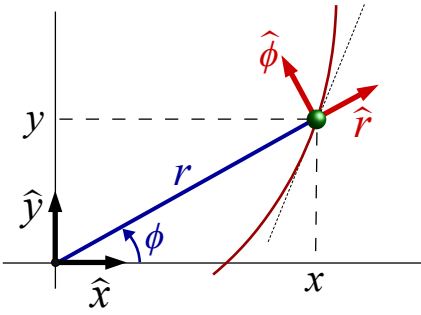
The acceleration is the rate of change of velocity:

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{dv_x}{dt} \hat{x} + \frac{dv_y}{dt} \hat{y} \quad (\text{A-2})$$

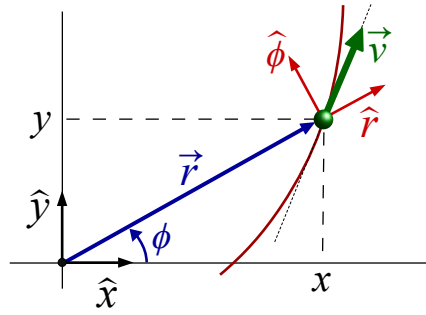
We can also write the acceleration in terms of the position:

$$\vec{a} = \frac{d^2\vec{r}}{dt^2} = \frac{d^2x}{dt^2} \hat{x} + \frac{d^2y}{dt^2} \hat{y} \quad (\text{A-3})$$

The acceleration has x -component $a_x = \ddot{x} = \dot{v}_x$ and y -component $a_y = \ddot{y} = \dot{v}_y$. In the *Principia*, Newton often says “velocity” when he is referring to speed and he often says “force” when he is referring to acceleration.



Unit Vectors



Position and Velocity

Velocity and Acceleration in polar coordinates

Much of the *Principia* is about central forces and the orbits they produce, and that is a topic best handled in polar coordinates.

The position vector $\vec{r} = r \hat{r}$ points from the origin to the object's location, and its magnitude can be a function of time $r(t)$ or angle $r(\phi)$. The velocity is the rate of change of position:

$$\vec{v} = \frac{d\vec{r}}{dt} = \dot{r} \hat{r} + r \dot{\phi} \hat{\phi} \quad (\text{A-4})$$

The radial component of the velocity is $v_r = \dot{r}$ and the azimuthal component is $v_\phi = r \dot{\phi}$. The radial component is the rate of change of the position vector's magnitude and the azimuthal component is the rate at which $\vec{r}$ changes direction. The velocity vector always points in the tangential direction (which is the $\hat{\phi}$ direction *only* at a radial extremum when the radial component is zero $\dot{r} = 0$).

The acceleration is the rate of change of velocity:

$$\vec{a} = \frac{d\vec{v}}{dt} = (\ddot{r} - r \dot{\phi}^2) \hat{r} + (r \ddot{\phi} + 2\dot{r} \dot{\phi}) \hat{\phi} \quad (\text{A-5})$$

The radial component of the acceleration is $a_r = \ddot{r} - r \dot{\phi}^2$ and the azimuthal component is $a_\phi = r \ddot{\phi} + 2\dot{r} \dot{\phi}$. The $r \dot{\phi}^2$ term is the centripetal acceleration due to the object's angular motion and the $2\dot{r} \dot{\phi}$ term is the Coriolis acceleration.

Newton's Second Law

We will make frequent use of the modern version of Newton's 2nd Law to solve a variety of problems in the *Principia*.

$$m \frac{d^2 \vec{r}}{dt^2} = \vec{F} \quad (2-1)$$

An object's mass times its acceleration equals the net force acting on it.

Constant Forces

For constant forces, the 2nd Law takes a familiar form.

$$m \vec{a} = \vec{F}$$

Finding the object's acceleration, velocity, and position requires only algebra when the net force $\vec{F}$ is constant, but we need to use calculus when the net force is a function of time, velocity, or position.

Time-dependent force

When the net force is a function of time, we use the definition of velocity to write the 2nd Law as

$$m \frac{dv}{dt} = F(t) \quad (2-2)$$

where $F(t)$ is the time-dependent net force. We integrate this equation find the velocity as a function of time.

$$v(t) = v_0 + \frac{1}{m} \int_0^t F(t) dt \quad (2-3)$$

To calculate the position as a function of time, we integrate again.

$$x(t) = x_0 + \int_0^t v(t) dt \quad (2-4)$$

These reduce to Galileo's kinematic expressions when the acceleration is constant.

Velocity-dependent force

When the net force is a function of the velocity, we write the 2nd Law as

$$m \frac{dv}{dt} = F(v) \quad (2-5)$$

where $F(v)$ is the velocity-dependent net force. We integrate this equation as well.

$$\int_{v_0}^v \frac{dv}{F(v)} = \frac{1}{m} t \quad (2-6)$$

This gives us time as a function of velocity, an expression we can solve for $v(t)$. Once we have the velocity as a function of time, we use Eq. (2-4) to calculate the position as a function of time.

Position-dependent force

When the net force is a function of position only, we write the Second Law as

$$m \frac{d^2 x}{dt^2} = F(x) \quad (2-7)$$

where $F(x)$ is the position-dependent net force. For some position-dependent forces this second-order ordinary differential equation has an analytic solution. The solution is the position as a function of time, and its derivative with respect to time is the velocity.

We integrate this equation to produce an alternative to the 2nd Law in terms of the position and time.

$$\frac{dx}{dt} = \pm \sqrt{\frac{2}{m} \int F(x) dx} \quad (2-8)$$

The choice of sign depends on the direction the object is moving.

Eq. (2-8) is solvable by direct integration for some functions $F(x)$.

$$\int_{x_0}^x \frac{dx}{\sqrt{\int_{x_0}^x F(x) dx}} = \pm \sqrt{\frac{2}{m}} t \quad (2-9)$$

This produces an equation we can solve for the position as a function of time if we can do the two integrals.

The chain rule provides a simple change of variable:

$$\frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = v \frac{dv}{dx}$$

This lets us write the 2nd Law for a position-dependent force in terms of the velocity and position.

$$mv \frac{dv}{dx} = F(x) \quad (2-10)$$

This is an equation we can integrate to get the velocity as a function of position.

$$v^2 - v_0^2 = \frac{2}{m} \int_{x_0}^x F(x) dx \quad (2-11)$$

This is equivalent to the modern Work-Energy Theorem in 1-dimension.

$$\frac{1}{2}mv^2 - \frac{1}{2}mv_0^2 = \int_{x_0}^x F(x) dx \quad (2-12)$$

The change in kinetic energy equals the total work done on the object.

Central Forces and Angular Momentum

Central forces play an important role in the *Principia* and Newton's derivation of Universal Gravitation. A **central force** always points in the radial direction, either toward or away from the force center, with a magnitude that depends only on the radial distance. Central forces have no azimuthal component.

In vector notation, the central force acting on an object with mass m is

$$\vec{F} \equiv -m f(r) \hat{r} \quad (\text{C-1})$$

where m is the object's mass and $f(r)$ is its acceleration due to the central force. For bound orbits, $f(r)$ is positive so the force always points toward the force center. Eqs. (A-5) and (C-1) together give us the radial and angular equations of motion for a central force.

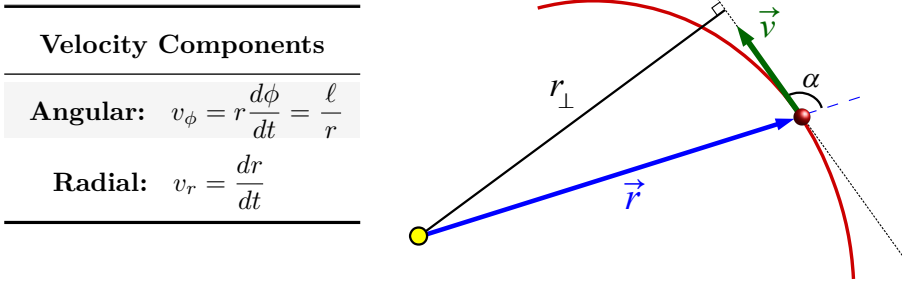
$$\ddot{r} - r\dot{\phi}^2 = -f(r) \quad (\text{C-2a})$$

$$r\ddot{\phi} + 2\dot{r}\dot{\phi} = 0 \quad (\text{C-2b})$$

We rewrite the angular component from Eq. (C-2b) as

$$\frac{1}{r} \frac{d}{dt} (r^2 \dot{\phi}) = 0 \quad (\text{C-3})$$

which defines a constant of motion ℓ called the **angular momentum** (per unit mass). The angular momentum vector $\vec{\ell} = \ell \hat{n}$ has magnitude $\ell = r^2 \dot{\phi}$ and its direction $\hat{n} = \hat{r} \times \hat{v}$ is perpendicular to the plane defined by $\vec{r}$ and $\vec{v}$.



An orbiting object's angular momentum also follows from its position and velocity.

$$\ell = |\vec{r} \times \vec{v}| = rv \sin \alpha = r_\perp v = r v_\phi \quad (\text{C-4})$$

The perpendicular distance from the origin to the line of the velocity is $r_\perp$ and the angle α is between $\vec{r}$ and $\vec{v}$ (see the above **figure**).

The angular momentum vector is conserved for central forces:

$$\frac{d\vec{\ell}}{dt} = \frac{d\vec{r}}{dt} \times \vec{v} + \vec{r} \times \frac{d\vec{v}}{dt} = \vec{v} \times \vec{v} + \vec{r} \times \vec{f} = 0$$

The $\vec{v} \times \vec{v}$ term is always zero of course, but the $\vec{r} \times \vec{f}$ term is zero *only* when the force is central.

The solutions to the two coupled differential equations for the orbit are the radial $r(t)$ and angular $\phi(t)$ positions as functions of time.

$$\frac{d^2 r}{dt^2} = -f(r) + \frac{\ell^2}{r^3} \quad (\text{C-5a})$$

$$\frac{d\phi}{dt} = \frac{\ell}{r^2} \quad (\text{C-5b})$$

This requires **four initial conditions** to completely specify the solution: We need the initial radial position r_0 , the initial angular position ϕ_0 , the angular momentum ℓ , and the initial radial speed v_0 .

Radial position: $r_0 = r(0)$

Angular position: $\phi_0 = \phi(0)$

Radial speed: $v_0 = \left. \frac{dr}{dt} \right|_{t=0}$ or $d_0 = \left. \frac{dr}{d\phi} \right|_{\phi=\phi_0}$

Angular momentum: ℓ

We combine the two orbital Eqs. (C-5) into a single equation for the orbit as a function of angle $r(\phi)$.

$$\frac{d^2 r}{d\phi^2} - \frac{2}{r} \left(\frac{dr}{d\phi} \right)^2 - r = -\frac{r^4}{\ell^2} f(r) \quad (\text{C-6})$$

The orbital solution $r(\phi)$ requires the initial positions $r_0 = r(\phi_0)$, the angular momentum ℓ , and the initial radial derivative d_0 as initial conditions.

It is sometimes more convenient to recast Eq. (C-6) in terms of the reciprocal variable $u \equiv 1/r$.

$$\frac{d^2 u}{d\phi^2} + u = \frac{1}{u^2 \ell^2} f(u) \quad (\text{C-7})$$

This important and useful equation is equivalent to Eq. (C-6) but it is mathematically simpler and it has more known exact solutions.

In BOOK I of the *Principia*, Newton presents various solutions to two fundamental problems in physics and astronomy. We will make good use of Eq. (C-6) and **Eq. (C-7)** as we offer our solution to these same problems.

The force-to-orbit problem: If you know the central force $f(r)$, solve the ordinary differential Eq. (C-6) or **Eq. (C-7)** and apply the appropriate initial conditions to calculate the orbit $r(\phi)$.

The orbit-to-force problem: If you know the orbit $r(\phi)$, then use Eq. (C-6) or **Eq. (C-7)** and take the appropriate derivatives to find the central force $f(r)$.

Energy Conservation

Energy Conservation is one of the fundamental principles of classical mechanics. The Law of Energy Conservation says the total energy of an isolated system is constant. Energy can neither be created nor destroyed; it can only be transformed from one form to another or transferred from one object to another.

The energy concept as we know it today was not available to Newton. Nonetheless, he did occasionally use the equivalent of the integral of the equation of motion Eq. (C-5a) that defines the modern concept of Energy Conservation.

$$\frac{1}{2} \left(\frac{dr}{dt} \right)^2 + V(r) + \frac{1}{2} \frac{\ell^2}{r^2} = E \quad (\text{E-1})$$

The mathematical integration constant is the physical conserved energy E (per unit mass), and we define the potential energy per unit mass $V(r)$.

$$V(r) = -\frac{1}{m} \int \vec{F}(r) \cdot d\vec{r} = + \int f(r) dr \quad (\text{E-2})$$

This unorthodox definition in terms of $f(r)$ without a minus sign is due to the minus sign in our definition of a central force from Eq. (C-1).

For central forces, we can rewrite the velocity's angular component in terms of the conserved angular momentum.

$$v^2 = \left(\frac{dr}{dt} \right)^2 + \frac{\ell^2}{r^2} \quad (\text{E-3})$$

This lets us express Energy Conservation in a simpler form in terms of the magnitude v of the velocity vector.

$$\frac{1}{2} v^2 + V(r) = E \quad (\text{E-4})$$

Modern orbital analysis for central forces sometimes parses Eq. (E-1) differently by including all the radial dependence in the **effective potential**.

$$V_{\text{eff}}(r) \equiv V(r) + \frac{1}{2} \frac{\ell^2}{r^2} \quad (\text{E-5})$$

Then energy conservation (per unit mass) takes a different form.

$$\frac{1}{2} \left(\frac{dr}{dt} \right)^2 + V_{\text{eff}}(r) = E \quad (\text{E-6})$$

This form is particularly useful when we calculate the maximum and minimum orbital distances where the radial speed is zero ($\dot{r} = 0 \Rightarrow V_{\text{eff}}(r_{\text{ext}}) = E$).

Physics in 1684: What Newton Knew

Newton's work in the *Principia* in the late 17th century was revolutionary and ground-breaking, but he did not start from scratch. In a famous quote from a 1675 letter to Robert Hooke "If I have seen further than others, it is by standing upon the shoulders of giants" Newton acknowledged the accomplishments of his predecessors. His metaphorical quote suggests he understood that his discoveries about mathematics and the laws of physics owed a debt to the work of other scientists.

Nicolaus Copernicus (1473–1543) published his heliocentric theory as he lay on his death bed, a theory that moved the Earth's position from the center of the universe to the third rock from the Sun. In the early 1600s, Johannes Kepler discovered his three laws of planetary motion, showing the planets orbit the Sun in elliptical orbits with varying speeds and distances from the Sun. Around the same time, Galileo Galilei used controlled experiments to deduce precise mathematical relationships between elapsed time and distance, velocity, and acceleration, for uniform and uniformly accelerated motion.

René Descartes (1596–1650) published *Principles of Philosophy* in 1644 where he proposed a mechanical view of the universe that objects can interact only if they make physical contact. To explain celestial motion, he devised a vortex theory where space consists of a rotating fluid and vortices in that fluid carry the planets and comets in their orbits. Proponents of this mechanical model criticized Newton for introducing forces that acted at a distance without any physical contact and Newton used a significant part of the *Principia* to refute this model. To emphasize his differences with Descartes, Newton went so far as to give his book a title (*Mathematical Principles of Natural Philosophy*) that clearly referenced the earlier work of Descartes, in part to show he was attempting to replace the erroneous Cartesian system with his own.

Christiaan Huygens (1629–1695) worked on the problem of circular motion and in 1673 he published the now standard formula for the net force exerted on an object moving in circular motion at constant speed. Huygens mistakenly thought circular motion is due to a balanced centrifugal force, but most of his results are still valid if we interpret his centrifugal force as Newton's centripetal force.

Several of Newton's predecessors produced results that are particularly relevant to his work in the *Principia* and they are worthy of a more detailed look.

Galileo: Motion with Constant Acceleration

Galileo Galilei (1564–1642) is one of the most important figures in the history of science. He made original, pioneering contributions to physics, astronomy, and the scientific method. He was among the very first to use science to answer questions about nature and as his famous quote "the book of nature is written in mathematical language" suggests, he answered them mathematically.

Galileo designed experiments to answer specific questions and he used mathematics to report his results. He asked and answered three of the most important and interesting questions in the history of physics and astronomy.

1. Astronomy: *Is the Sun the center of the solar system?*

Galileo did not invent the telescope but he did design and build his own telescopes with significantly increased magnification and turned them into tools of scientific observation. In an attempt to decide which astronomical theory correctly describes the universe, Galileo aimed his telescope upward and made six observations that told him the answer.

1. Venus goes through phases like the Moon
2. Jupiter has four permanent moons
3. the Sun has spots that change and move
4. the Sun rotates on its axis
5. the Moon has mountains and craters
6. the Milky Way is very far away

These observations all support the heliocentric model of Copernicus and they all contradict Aristotle's physics and his geocentric model. Galileo published his early observations in 1610 in a short book called *Sidereal Messenger*. He later published a strong defense of the heliocentric system in *Dialogue Concerning the Two Chief World Systems* (1632) which precipitated his conflict with religious authorities.

2. Kinematics and Free Fall: *Do heavy objects fall faster than light ones?*

In 1638, less than 4 years before his death, Galileo published *The Discourses and Mathematical Demonstrations Relating to Two New Sciences* (usually referred to as *Two New Sciences*). This was Galileo's final book where he summarized his work in physics over the preceding thirty years. [24]

Using an inclined plane, Galileo performed experiments on uniformly accelerated motion. The modern equations of 1-dimensional kinematics summarize and confirm Galileo's experimental results for motion with a constant acceleration.

$$\Delta x = v_{\text{ave}} \Delta t = v_0 \Delta t + \frac{1}{2} a \Delta t^2 \quad (\text{G-1a})$$

$$v = v_0 + a \Delta t \quad (\text{G-1b})$$

$$v^2 = v_0^2 + 2a \Delta x \quad (\text{G-1c})$$

The change in velocity is proportional to the time interval, and for an object released from rest the distance travelled is proportional to the time-squared and the velocity squared. When the acceleration is zero, the object moves in uniform motion at constant velocity and the distance is proportional to the time.

The current historical consensus is that Galileo never actually did the famous free-fall experiment where he simultaneously dropped two cannonballs with different masses off the Tower of Pisa. It was probably a thought experiment that he used to argue against Aristotle's incorrect claim that objects fall at a rate that

is proportional to their mass. Aristotle was a brilliant man who made significant contributions to the natural sciences, philosophy, and politics, but apparently the man never dropped anything.

We do know that Galileo did many easier, more conclusive experiments with inclined planes and he predicted the correct result: The acceleration due to gravity does **not** depend on the mass of the falling object. If we can minimize the effects of air resistance, the two falling balls would land at the same time. This was confirmed in August 1971 by Apollo 15 astronaut when the hammer and feather he dropped on the surface of the Moon landed simultaneously. [25]

3. Projectile Motion: *What is the shape of a projectile's trajectory?*

Galileo used the same experimental approach to study projectile motion by using inclined planes to launch projectiles horizontally. He demonstrated that a projectile experiences two independent motions: The horizontal motion is uniform and the vertical motion is uniformly accelerated by gravity.

Projectile Motion is an example of 2-d motion under a constant force where the acceleration is $\vec{a} = -g\hat{y}$ if we ignore air resistance and the effects of Earth's size and rotation. Substituting this acceleration into Eqs. (G-1) gives us the projectile's horizontal and vertical positions as functions of time.

$$x = x_0 + v_{0x}t \quad (\text{G-2a})$$

$$= x_0 + v_0 t \cos \theta_0$$

$$y = y_0 + v_{0y}t - \frac{1}{2}gt^2 \quad (\text{G-2b})$$

$$= y_0 + v_0 t \sin \theta_0 - \frac{1}{2}gt^2$$

The initial velocity's components depend on the projectile's initial speed v_0 and launch angle θ_0 .

We use Eqs. (G-2) to eliminate time and find the projectile's trajectory (its height y as a function of its range x).

$$y = y_0 + x \tan \theta_0 - \frac{g}{2v_0^2 \cos^2 \theta_0} x^2 \quad (\text{G-3})$$

The trajectory is a parabola! When the initial velocity is small enough that we can ignore the effects of air resistance, the projectile follows a parabolic trajectory.

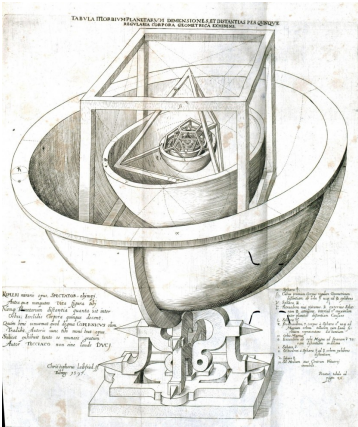
The projectile's horizontal speed is constant so the change in its horizontal position Δx is constant during equal time intervals. The vertical motion is accelerated so the change in the projectile's height Δy is not constant during equal time intervals.

Newton mentions or uses Galileo's results frequently in the *Principia* (in COROLLARY 1 to LEMMA 10 and the SCHOLIUM to PROPOSITION 10 in BOOK I to mention just two). Galileo's results are simple and useful, and we still teach and use them today, but they are empirical: They describe but do not explain the motion. As we will see, Newton used his Laws of Motion to explain Galileo's results theoretically.

Kepler: Laws of Planetary Motion

Johannes Kepler (1571–1630) was another important contributor to the scientific revolution. His 1596 book *The Cosmographic Mystery* was a defense of the heliocentric model that included what he considered to be a key insight into the structure of the solar system. He proposed a **geometric explanation** for the distances between the six known planets by fitting the five Platonic solids in the five gaps between six spherical shells that represented the planetary orbits.

Saturn	
	Cube
Jupiter	
	Tetrahedron
Mars	
	Dodecahedron
Earth	
	Icosahedron
Venus	
	Octahedron
Mercury	



Kepler’s numerically-inspired model of the solar system.

This numerical coincidence has no physical significance, but fortuitously for Kepler two good things came of it. Tyco Brahe read Kepler’s book and gave him a job that ultimately lead to access to Brahe’s excellent observational data of planetary positions. Second, he had to adjust the thickness of the planetary spheres and this was the first clue that the planetary orbits are not circles.

After Brahe’s death in 1601, Kepler continued the detailed and painstaking data analysis that took over 20 years and he discovered three laws of planetary motion. His book *A New Astronomy* (1609) included his first two laws and the final section of *The Harmony of the World* (1619) involved his discovery of his third law.

Kepler’s 1st Law: The Ellipse Law

The orbit of every planet is an ellipse with the Sun at one focus.

Kepler’s 2nd Law: The Area Law

A line joining a planet and the Sun sweeps out equal areas in equal times.

Kepler’s 3rd Law: The Period Law

The square of a planet’s orbital period is directly proportional to the cube of its average distance from the Sun.

Kepler’s 2nd Law is valid for any central force, but the 1st and 3rd Laws apply only to the inverse-square central force.

Kepler's 1st Law: The Ellipse Law

The geometric expression for Kepler's off-center **elliptical orbit** is

$$r(\phi) = \frac{b^2/a}{1 - \varepsilon \cos \phi} \quad (\text{K-1})$$

where a is the ellipse's semi-major axis and $b = a\sqrt{1 - \varepsilon^2}$ is its semi-minor axis. The eccentricity ε is a measure of how much the orbit differs from a circle. Circular orbits have zero eccentricity and free-fall orbits ($\ell = 0$) have the maximum $\varepsilon = 1$.

For the attractive inverse-square central force $f(r) = \beta/r^2$, the dynamic expression for Kepler's elliptical orbit depends on the angular momentum ℓ , the force constant β , and the initial radial position r_0 .

$$r(\phi) = \frac{\ell^2/\beta}{1 - \left(1 - \frac{\ell^2/\beta}{r_0}\right) \cos \phi} \quad (\text{K-2})$$

The orbit is completely determined by the ratio ℓ^2/β and the initial conditions.

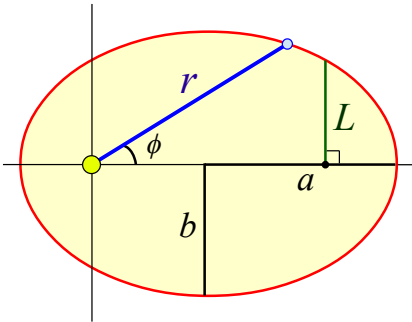
We write the eccentricity in terms of the geometric or dynamic parameters.

$$\varepsilon = \sqrt{1 - \frac{b^2}{a^2}} = 1 - \frac{\ell^2}{\beta r_0} \quad (\text{K-3})$$

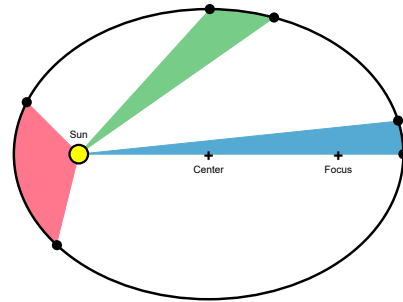
The orbit is simpler in terms of the initial position r_0 or the semi-major axis a .

$$r(\phi) = \frac{r_0(1 - \varepsilon)}{1 - \varepsilon \cos \phi} = \frac{a(1 - \varepsilon^2)}{1 - \varepsilon \cos \phi} \quad (\text{K-4})$$

For this choice of coordinate system, the maximum distance is the initial distance, and the minimum distance occurs at $\phi = \pi$. The average radial distance is the average of these two extremes, and it equals the semi-major axis a .



The Ellipse Law



The Area Law [26]

The **semi-latus rectum** L is the chord from one focus to the ellipse that is perpendicular to the major axis.

$$L = \frac{b^2}{a} = a(1 - \varepsilon^2) \quad (\text{K-5})$$

Newton frequently refers to the semi-latus rectum in the *Principia*.

Kepler's 2nd Law: The Area Law

As an object moves along an orbit under the influence of any central force, its radial position vector sweeps out an area that is approximately triangular. We will see in PROPOSITION 1 of BOOK I that a little geometry, some calculus, and the definition of angular momentum lead us directly to Kepler's 2nd Law.

$$\Delta A = \frac{1}{2} \ell \Delta t \quad (\text{K-6})$$

The area of a “slice” of the orbit is proportional to the elapsed time so the radius vector sweeps equal areas in equal times. For periodic orbits, we apply Kepler's 2nd Law to the entire orbit and relate the enclosed area to the orbital period.

$$A = \frac{1}{2} \ell T \quad (\text{K-7})$$

The Area Law is a consequence of angular momentum conservation, so it is valid for any central force. From the very first proposition of the *Principia*, Newton uses the Area Law (with no mention of Kepler) to relate orbital areas to elapsed times.

Kepler's 3rd Law: The Period Law

We use the definitions of angular momentum from Eq. (C-3) and eccentricity from Eq. (K-3) to calculate the period of an elliptical orbit. [27]

$$T = \frac{1}{\ell} \int_0^{2\pi} r^2(\phi) d\phi = \sqrt{\frac{a^3}{GM_\odot}} (1 - \varepsilon^2)^{3/2} \int_0^{2\pi} \frac{d\phi}{(1 - \varepsilon \cos \phi)^2} \quad (\text{K-8})$$

All the dependence on the eccentricity cancels and the result is Kepler's 3rd Law.

$$T^2 = \frac{4\pi^2}{GM_\odot} a^3 \quad (\text{K-9})$$

The period is related to the average size of the elliptical orbit in a very simple way that follows from Newton's 2nd Law and Universal Gravitation. The proportionality constant depends on the gravitation constant G , the mass $M_\odot$ of the Sun, and the planet's average distance a . In the *Principia*, Newton not only provides Kepler's proportionality constant, he also adds a center-of-mass correction which depends on the orbiting planet's mass m .

$$T^2 = \frac{4\pi^2}{GM_\odot \left(1 + \frac{m}{M_\odot}\right)} a^3 \quad (\text{K-10})$$

The value of the force constant $GM_\odot = 4\pi^2 \text{ au}^3/\text{y}^2$ for the Earth-Sun system is accurate to approximately one part in 332 900. [28]

Kepler's Laws are simple and useful, and we still teach and use them today, but they are empirical. Even though they are based on observational data, they only describe the planetary orbits; they do not explain them. Newton's Laws of Motion and Universal Gravitation explain Kepler's Laws theoretically. This agreement between theory and observation is convincing evidence that Newton was right.

Huygens: Uniform Circular Motion

Christiaan Huygens (1629–1695) was a contemporary of Newton who made significant contributions in physics and astronomy. He published his classic book *Horologium Oscillatorium* (*The Pendulum Clock*) in 1673 (14 years before the first edition of the *Principia*) where he described his theory of pendulum motion, and he explained how he built accurate pendulum clocks that let him accurately measure the acceleration due to earth’s gravity.

When Richard Bentley asked Newton in 1691 how to prepare to read the *Principia*, Newton recommended a daunting list of mathematical and scientific books, but he singled out one in particular: the *Horologium Oscillatorium*. Newton owned a copy, he was familiar with its contents, and he greatly admired it. [29]

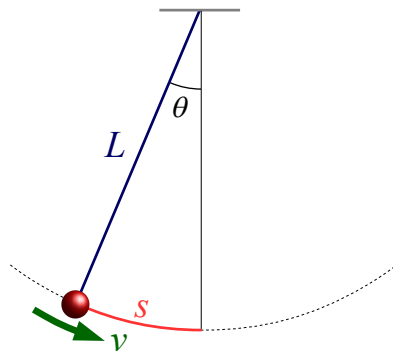
Part II of the *Horologium* is entitled “The falling of heavy bodies and their motion in a cycloid” and opens with three “hypotheses” of motion that codify Galileo’s experimental results on projectile motion. The first hypothesis is a less-general predecessor of Newton’s 1st Law that gives a restricted definition of inertia. The other two hypotheses concern the principle of composition of motion in free fall and the independence of the horizontal and vertical components of projectile motion. [30]

Part IV of the *Horologium* is entitled “Concerning the center of oscillation” and ends with a proposition that tells us how “to find the distance a heavy body falls vertically in a given time” using the length of a **seconds pendulum**.

A **simple pendulum** consists of an object swinging back and forth while suspended from a string of negligible mass. For small oscillations, the component of the net force along the object’s motion is approximately proportional to the arc length and produces simple harmonic motion (SHM).

A **seconds pendulum** is a simple pendulum with a 2-second period, one second for the swing in one direction and one second for the return swing, so its frequency is $\frac{1}{2}$ Hz.

$$T = 2 \text{ s} \iff \text{frequency} = \frac{1}{2} \text{ Hz}$$



Huygens used his theory of pendulum motion to relate the distance d_1 of a one-second free-fall from rest to the length L_1 of a seconds pendulum that is easier to measure accurately. For small oscillations, the period of simple pendulum motion is $T = 2\pi\sqrt{L/g}$ so the length of a seconds pendulum is $L_1 = g/\pi^2$. The distance an object falls from rest is $d = \frac{1}{2}gt^2$ so $d_1 = \frac{1}{2}g$ is the distance fallen in the first second. The pendulum length is related to the free-fall distance $d_1 = \frac{1}{2}\pi^2 L_1$ and the free-fall distance is related to gravity $g = 2d_1$. [31]

Huygens measured 15.096 Paris feet ($d_1 = 16.088$ ft) for a one-second fall, which corresponds to a value for the acceleration due to gravity that is less than $\frac{1}{40}$ percent smaller than the modern value for g at Paris. [31]

Huygens proposed using the length of a seconds pendulums to define a universal standard of length, a distance that is less than a centimeter smaller than our modern meter. His pendulum clock gave him the means to determine the length of a seconds pendulum accurately.

In Part V of the *Horologium*, Huygens discusses his construction of a clock using a conical pendulum and then he presents without proof thirteen theorems concerning his “centrifugal force” and circular motion. Six theorems (1–5, 13) are about Uniform Circular Motion, number 6 is about circular motion on the inner surface of a paraboloid, and the other six are about conical pendulums. [32]

Uniform Circular Motion (UCM) is motion at constant speed on a circular orbit. The velocity has a constant magnitude but changes direction so there is an acceleration — a **centripetal** acceleration. The magnitude of the acceleration is the square of the object’s speed divided by the circle’s radius $a_c = v^2/R$. The direction of the acceleration is toward the center of the circle.

UCM occurs in the plane defined by the velocity and acceleration vectors. Any component of the acceleration along the direction of the motion would change the object’s speed, so the acceleration must be perpendicular to the velocity.

Huygens thought circular motion is an equilibrium between some external constraint and an innate centrifugal tendency he called a force. Even so, his theorems remain valid if we replace his word “centrifugal” with Newton’s word “centripetal” and its modern connotations.

Theorem 1: The period of UCM is proportional to the radius.

The period $T = 2\pi R/v$ is the circumference divided by the speed so it is proportional to the radius for various circular orbits at a given constant speed.

Theorem 2: The centripetal force is inversely proportional to the radius.

Theorem 3: The centripetal force is proportional to the speed-squared.

The net force is the object’s mass times its centripetal acceleration $F_c = mv^2/R$.

Using this result, Huygens, Hooke, Halley and Wren derived the inverse-square law of gravitational attraction for the special case of circular orbits that Newton discusses in COROLLARY 6 to PROPOSITION 4 in BOOK I of the *Principia*.

Theorem 4: If two equal masses moving in circles with different radii experience the same centripetal force, then the ratio of the periods equals the ratio of the square-root of the radii.

For equal forces, the speed is proportional to the square-root of the radius

$$\frac{v_1^2}{R_1} = \frac{v_2^2}{R_2} \implies \frac{v_2}{v_1} = \sqrt{\frac{R_2}{R_1}} \implies \frac{T_1}{T_2} = \frac{R_1/v_1}{R_2/v_2} = \sqrt{\frac{R_1}{R_2}} \quad (\text{H-1})$$

and the theorem follows from the definition of period.

Theorem 5: A small object moving along a circle at the speed acquired by falling from a height equal to half the radius will experience a centripetal force equal to its own weight.

Galileo's kinematics tell us the final speed of an object that is dropped from rest and falls a distance y is $v^2 = 2gy$. When that distance is $y = \frac{1}{2}R$ the final speed is $v^2 = Rg$ and the net force $mv^2/R = mg$ equals the object's weight.

Theorem 6: All horizontal circular orbits on the inner surface of a paraboloid with a vertical axis have the same period that equals the period of the pendulum whose length is equal to the semi-latus rectum of the generating parabola.

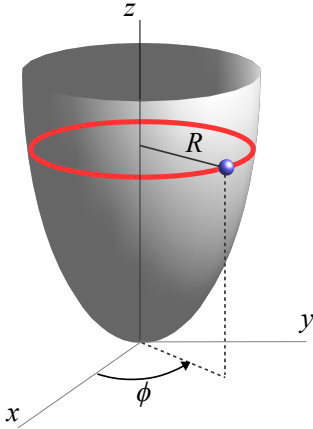
The generating parabola is $y = x^2/(2L)$ where L is the parabola's semi-latus rectum. The resulting paraboloid after revolution about the z axis is $z = r^2/(2L)$. We can derive the equation of motion from the Lagrangian in cylindrical coordinates.

$$\mathcal{L} = \frac{1}{2}m\dot{r}^2 \left(1 + \frac{r^2}{L^2}\right) + \frac{1}{2}mr^2\dot{\phi}^2 - \frac{mg}{2L}r^2 \quad (\text{H-2})$$

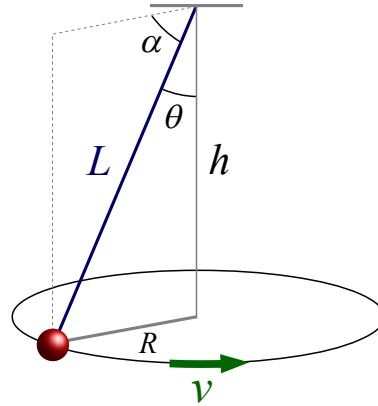
After the standard Lagrangian shenanigans, we get the equation for motion on the paraboloid's surface.

$$\left(1 + \frac{r^2}{L^2}\right)\ddot{r} + \frac{r}{L^2}\dot{r}^2 = r\dot{\phi}^2 - \frac{g}{L}r \quad (\text{H-3})$$

The left-hand side is zero for circular motion ($\ddot{r} = \dot{r} = 0$) and $r\dot{\phi} = v$ for UCM, so the speed along a horizontal circle on a paraboloid is $v = R\sqrt{g/L}$ and the period is $T = 2\pi\sqrt{L/g}$. This is the same period as a pendulum of length L and the period for a horizontal circle is independent of the circle's radius.



Theorem 6: A Paraboloid



Theorems 7-12: A Conical Pendulum

Theorem 7: Two conical pendulums with the same height have the same period, no matter the length of their strings.

Theorem 8: The period of a conical pendulum is proportional to the square root of the pendulum's height.

A **conical pendulum** consists of an object moving in uniform circular motion attached to a string that traces out a cone. It is a simple pendulum with a string of length L where the object moves in a horizontal circle of radius R instead of swinging back and forth. The cone's height is related to the constant apex angle θ .

$$h = \frac{R}{\tan \theta} = L \cos \theta$$

The 2nd Law analysis of a conical pendulum relates the net force and weight to the tension F_T in the string and the angle.

$$m \frac{v^2}{R} = F_T \sin \theta \quad (\text{H-4a})$$

$$mg = F_T \cos \theta \quad (\text{H-4b})$$

Unlike a simple pendulum, there is no component of the acceleration in the direction of the object's motion so the speed is constant. The speed is $v = \sqrt{Rg \tan \theta}$ so the period is $T = 2\pi\sqrt{h/g}$. The period is independent of the string's length and proportional to the square root of the pendulum's height.

Theorem 9: The period of a conical pendulum is π times the time to free-fall vertically from twice the pendulum's height h .

An object falls a distance h from rest in time $t_2 = 2\sqrt{h/g}$ so the pendulum's period $T = 2\pi\sqrt{h/g}$ is π times the free-fall time.

Theorem 10: When the period of the circular motion of a conical pendulum has the same period as a simple pendulum whose length equals the circle's radius, then the object experiences a centripetal force equal to its weight.

The period of the simple pendulum is $T = 2\pi\sqrt{R/g}$ so $v^2 = Rg$ and $mv^2/R = mg$.

This result also follows from the 2nd Law analysis of Eqs. (H-4) which tells us

$$\tan \theta = \frac{mv^2/R}{mg} \quad (\text{H-5})$$

so the centripetal force equals the object's weight at $\theta = 45^\circ$ and that is the angle where the conical pendulum's height equals the circle's radius.

Theorem 11: The period of a conical pendulum equals the time to fall vertically from a distance equal to the string's length when the angle the string makes to the horizontal is approximately 2.9° .

The conical pendulum's height is $h = L \sin \alpha$ where α is the angle the string makes with the horizontal, so its period is $2\pi\sqrt{L \sin \alpha/g}$. The time to free-fall a distance L from rest is $\sqrt{2L/g}$. The two times are equal when $\sin \alpha = 1/(2\pi^2)$ so the angle $\alpha \approx 2.904^\circ$ is very close to the given approximate value.

Huygens tells us the result is exact when $\sin \alpha$ equals the area of a square inscribed in a circle divided by the square of the circumference of that circle.

$$\sin \alpha = \left(\frac{\text{square side}}{\text{circumference}} \right)^2 = \left(\frac{\sqrt{2}R}{2\pi R} \right)^2 = \frac{1}{2\pi^2} \quad (\text{H-6})$$

This is a geometric expression for the equal-time condition.

Theorem 12: If two conical pendulums with objects of equal mass and strings of different lengths rotate so that the objects are at the same height, then the ratio of the tensions in the strings equals the ratio of the lengths of the strings.

The two objects are at the same height so geometry tells us $L_1 \cos \theta_1 = L_2 \cos \theta_2$. The two objects have the same mass so Eq. (H-4b) tells us $F_1 \cos \theta_1 = F_2 \cos \theta_2$ and Huygens's theorem follows.

$$\frac{\cos \theta_1}{\cos \theta_2} = \frac{F_2}{F_1} = \frac{L_2}{L_1} \quad (\text{H-7})$$

The tension in the string is $F_T = mg / \cos \theta = mg L / h$ so we can do Huygens one better. The tensions and lengths also have the same ratio when the two pendulums have the same mass-to-height ratio.

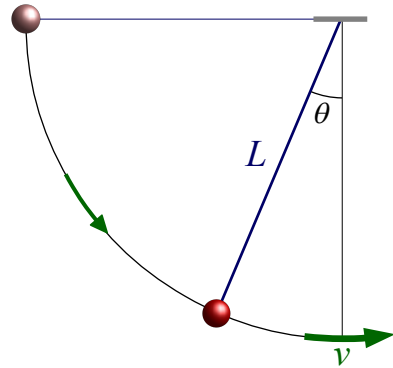
Theorem 13: If a **simple pendulum** starts from rest at the maximum height and swings through the whole 90° arc, when it arrives at its lowest point the tension in the string is three times the tension when it is simply suspended.

The object's initial potential energy equals its kinetic energy at its lowest point. Newton's 2nd Law relates the centripetal force to the tension and weight at the lowest point. Energy conservation tells us the speed at the lowest point is $v^2 = 2gL$ where the string length L is the radius of the circular path. Newton's 2nd Law tells us the tension at the bottom is the object's weight plus the net force.

Energy Conservation: $mgL = \frac{1}{2}mv^2$

Newton's 2nd Law: $\frac{mv^2}{L} = F_T - mg$

Tension: $F_T = mg + m\frac{v^2}{L} = 3mg$



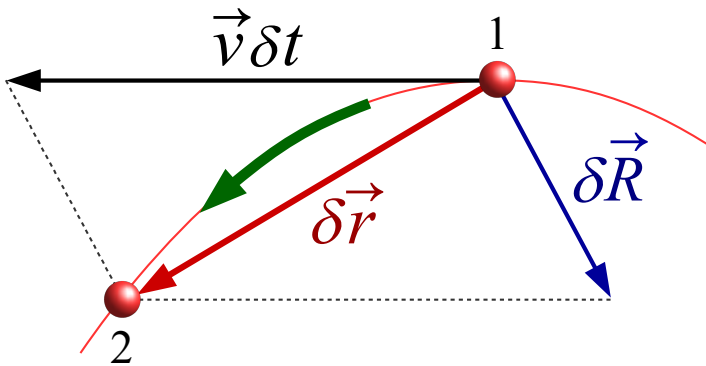
The tension in the string at the lowest point of the swing is three times the object's weight. When the object is suspended motionless, the tension in the string equals the object's weight.

Hooke: Resolving Orbital Motion

Robert Hooke (1635–1703) was a contemporary of Newton who did important original work in many different fields including physics, mathematics, architecture, and biology. Early in his career, he was an assistant to Robert Boyle and he built the apparatus and performed the experiments that led to the discovery what we now call Boyle’s Law relating the pressure and volume of a gas. His 1665 book *Micrographia* is one the great 17th century science books. It was the first book to illustrate insects and other small phenomena (including the first sighting of biological cells) as seen through a microscope and it made Hooke famous. His observations and exquisite drawings showed what a powerful tool the microscope could be, doing for the microscope and biology what Galileo did for the telescope and astronomy.

His long-running quarrel with Newton started in 1672 when he publicly criticized Newton’s recently published theory on optics. That did not stop Hooke, in his new role as Secretary of the Royal Society, from writing to Newton in 1679 in the hope he could entice Newton to submit new material the Royal Society could publish. [1]

In his first letter, Hooke listed several topics for Newton’s consideration, including what we now call Hooke’s Law for the force exerted by a spring, and Hooke’s hypothesis that “compounding the celestial motions of the planets of a direct motion by the tangent and an attractive motion towards the central body” is a useful way to analyze orbital motion.



Hooke’s Hypothesis

In modern terminology, Hooke was suggesting a way to analyze orbital motion by resolving the change in position $\delta \vec{r}$ into its **tangential** and **radial** components. The tangential component $\vec{v}\delta t$ is inertial, straight-line motion in the direction of the velocity vector and the radial component $\delta \vec{R}$ is accelerated motion toward the force center. The two components are not generally perpendicular (although they are for circular orbits and at the radial extremes of an ellipse).

The key words in Hooke's hypothesis are planetary "motion" and the "attraction" towards the Sun. He has abandoned the "centrifugal tendency" of Descartes and Huygens for an attractive central force that, in Newton's capable hands, led to Universal Gravitation. This was more than a change in direction — it was a change in physics. [1]

Newton's final recognition of the centripetal direction of a gravitational force only came after he received Hooke's letter. Hooke's hypothesis essentially defined the attractive central force that Newton named "centripetal" and used throughout the *Principia* to great effect.

It is not clear whether Hooke's hypothesis immediately caused Newton to change from centrifugal to centripetal forces, but Hooke had reasonable grounds to demand some acknowledgment from Newton. Hooke produced a qualitative description of orbital mechanics in the mid-1660s that Newton probably read, and in 1674 he formulated a version of universal gravitational attraction. Newton later admitted that it was only after using Hooke's hypothesis that he discovered the Area Law held for any central force, and was then able to prove the inverse-square central force produces elliptical orbits. [33]

Hooke was never able to prove a mathematical connection between an inverse-square attractive force and Kepler's elliptical orbits. It is understandable that Hooke wanted recognition after reading BOOK I of the soon-to-be published *Principia*, but he went too far and accused Newton of stealing his idea of an inverse-square universal force of gravity. Newton's response was to delete nearly every mention of Hooke in the *Principia*. [34]

Alexis Clairaut's famous admonition that Hooke serves as an example of the great difference "between a truth that is glimpsed and a truth that is proven" seems harsh in hindsight. [4, page 21]

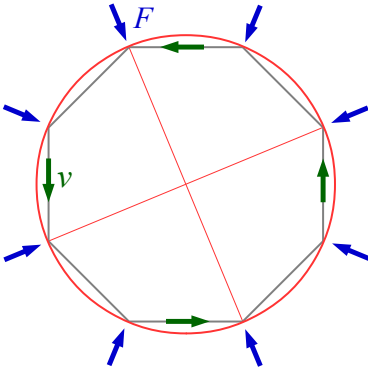
Hooke did not deserve as much credit as he demanded from Newton, but he deserves far more credit than history has given him for his many accomplishments.

Newton: Three pre-Principia approximations

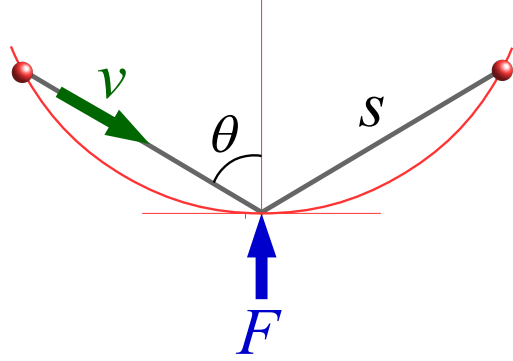
Newton's research on celestial orbits began in the 1660s (before, during, and after his "miracle year"). It is important to note that when he started his analyses, Newton was still thinking about circular motion in terms of a centrifugal tendency and not a centripetal force. By the end of the decade, he had produced three approximations that he used twenty years later when he did a more sophisticated analysis in the *Principia*. [35]

Polygonal Approximation

To analyze circular motion, Newton considered an object moving along a trajectory that follows the perimeter of a polygon inscribed in the circle. This approximates the circular orbit as a series of straight-line inertial motions with periodic radial impulses when the object collides elastically with the circle. This changes the direction of the object's momentum, but not the magnitude.



Polygonal Approximation



$$\text{Angle: } \theta = \frac{(n-2)}{2n} \pi = \frac{\pi}{2} - \frac{\pi}{n}$$

For a regular n -sided polygon, the sum of the vertex angles is $(n-2)\pi$ so each vertex angle is $\pi(n-2)/n$ and each side subtends an angle $2\pi/n$. When the polygon is inscribed in a circle of radius R , its perimeter is $2nR \sin(\pi/n)$ so the length of each side is $s = 2R \sin(\pi/n)$.

When the object collides with the circle, the change in its momentum is

$$\Delta \vec{p} = +2mv \cos \theta \hat{n} \quad (\text{N-1})$$

where the normal unit vector $\hat{n}$ points toward the center of the circle and the angle θ between the velocity and the normal is half the vertex angle. The time between collisions is the length of a side divided by the constant speed.

$$\Delta t = \frac{s}{v} = \frac{2R}{v} \sin \frac{\pi}{n} \quad (\text{N-2})$$

The net force is the rate of change of the object's momentum.

$$F = \lim_{\Delta t \rightarrow 0} \frac{\Delta p}{\Delta t} = \frac{mv^2}{R} \quad (\text{N-3})$$

This is the continuum limit where the number of sides increases to infinity ($n \rightarrow \infty$) so the polygon coincides with the circle, the velocity is tangential, and the force is continuous.

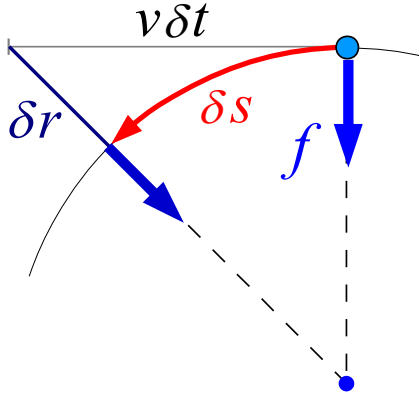
Newton mentions this method explicitly in the *Principia* in the SCHOLIUM after PROPOSITION 4 in BOOK I where he derives the v^2/R centripetal acceleration. He later extends this method to analyze any orbit using a smooth curve and a continuous force. The polygonal approximation does not provide enough information to solve the general orbit-to-force problem, so Newton had to devise other methods.

Parabolic Approximation

When an object moves under the influence of an attractive central force, it is deflected away from the tangential direction and toward the source of the force. Over a small time interval δt , the acceleration f due to the central force is approximately constant and it produces a small radial displacement δr that we can approximate.

$$\delta r \approx \frac{1}{2} f \delta t^2 \quad (\text{N-4})$$

This is essentially the same analysis Galileo used for projectile motion.



Parabolic Approximation

Newton will use this approximation often in the *Principia*, including PROPOSITION 6 of BOOK I where he presents his solution to the orbit-to-force problem that calculates the central force $f(r)$ from a known orbit $r(\phi)$.

Circular Approximation

In the 1660s, Newton found a method to calculate the force on an object moving along an elliptical orbit using a circle that is tangent to the orbit and has the same radius of curvature as the orbit at the tangent point. [35]

At the tangent point, the tangent circle has the same value, tangent line, and curvature as the orbit. The center of curvature of the orbit is the center of the tangent circle. Leibniz named this circle the **osculating circle** from the Latin word for a kiss. [20]

The radius of curvature of the orbit $r(\phi)$ at the point $\phi = \phi_k$ is

$$R_{\text{cur}} = \frac{\left(r^2 + (r')^2 \right)^{3/2}}{r^2 + 2(r')^2 - r r''} \bigg|_{\phi=\phi_k} \quad (\text{N-5})$$

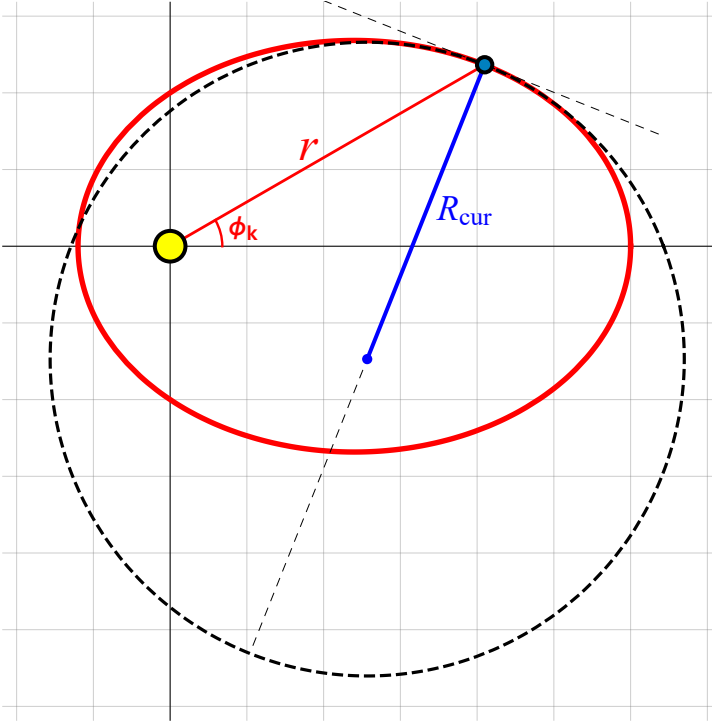
where r' and r'' are the first and second derivatives respectively of the orbit $r(\phi)$ with respect to the angle ϕ . [36]

The slope of the tangent line at that point is

$$S_{\tan} = \left. \frac{r' \sin \phi + r \cos \phi}{r' \cos \phi - r \sin \phi} \right|_{\phi=\phi_k} \quad (\text{N-6})$$

and the slope of the radial line perpendicular to the tangent is $S_{\text{rad}} = -1/S_{\tan}$.

The center of curvature is the point inside the orbit that is on the perpendicular line a distance R_{cur} from tangent point. The force on the object at the tangent point is approximately $F \approx mv^2/R_{\text{cur}}$ and the approximation is better for small or nearly circular arcs.



The **osculating circle** for an ellipse with $L = 1$ and $\varepsilon = \frac{2}{3}$ at the point $\phi_k = 30^\circ$.

Newton uses this method in COROLLARY 1 to PROPOSITION 6 in BOOK I where he presents his solution to the orbit-to-force problem.

It is now time for us to start our way through the *Principia*!